# SET NO: 20
## Stock Analysis

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**Course: DSA06012 Data Handling and Visualization**

## Question 1

Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.

### AIM

To visualise the price movement of Stock A, Stock B and Stock C over time using a line chart.

### PROCEDURE

1. Create the stock data frame and convert Date using as.Date().
2. Convert the data to long format with pivot_longer().
3. Plot geom_line() and geom_point() with colour = Stock and label the axes and title.

### CODE

```
# Dataset
library(ggplot2); library(dplyr); library(tidyr); library(gridExtra)
stock <- data.frame(
  Date    = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),
  Stock_A = c(100, 105, 110),
  Stock_B = c(150, 152, 148),
  Stock_C = c(120, 118, 122))
# (or) stock <- read.csv("Stock_Prices.csv"); stock$Date <- as.Date(stock$Date)

# Line chart of three stocks
long <- pivot_longer(stock, cols = starts_with("Stock"),
                     names_to = "Stock", values_to = "Price")
p1 <- ggplot(long, aes(x = Date, y = Price, colour = Stock)) +
  geom_line(linewidth = 1) + geom_point(size = 3) +
  labs(title = "Stock Prices of A, B and C (Jan 2023)",
       x = "Date", y = "Stock Price", colour = "Stock")
print(p1)
```

**OUTPUT**

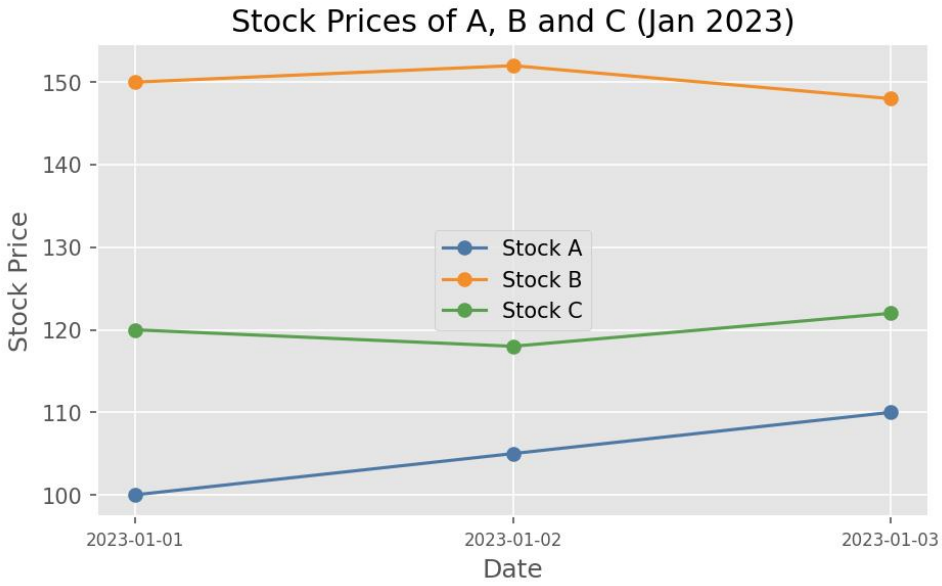



*Figure 20.1: Line chart of Stock A, B and C*

**RESULT**

Stock A rises steadily from 100 to 110, Stock B stays between 148 and 152, and Stock C fluctuates between 118 and 122.

## Question 2

Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.

**AIM**

To show the daily percentage change of Stock A using a bar chart.

**PROCEDURE**

1. Calculate the percentage change using diff() on the Stock A prices.
2. Plot geom_col() of the daily % change against Date.
3. Add value labels, title and axis labels.

**CODE**

```
# Daily percentage change for Stock A
stock$Pct_Change_A <- c(NA, diff(stock$Stock_A) / head(stock$Stock_A, -1) * 100)
print(stock)

p2 <- ggplot(na.omit(stock), aes(x = Date, y = Pct_Change_A)) +
  geom_col(fill = "steelblue", width = 0.4) +
  geom_text(aes(label = paste0(round(Pct_Change_A, 2), "%")), vjust = -0.5) +
  labs(title = "Stock A - Daily % Change",
       x = "Date", y = "Daily % Change")
print(p2)
```

**OUTPUT**

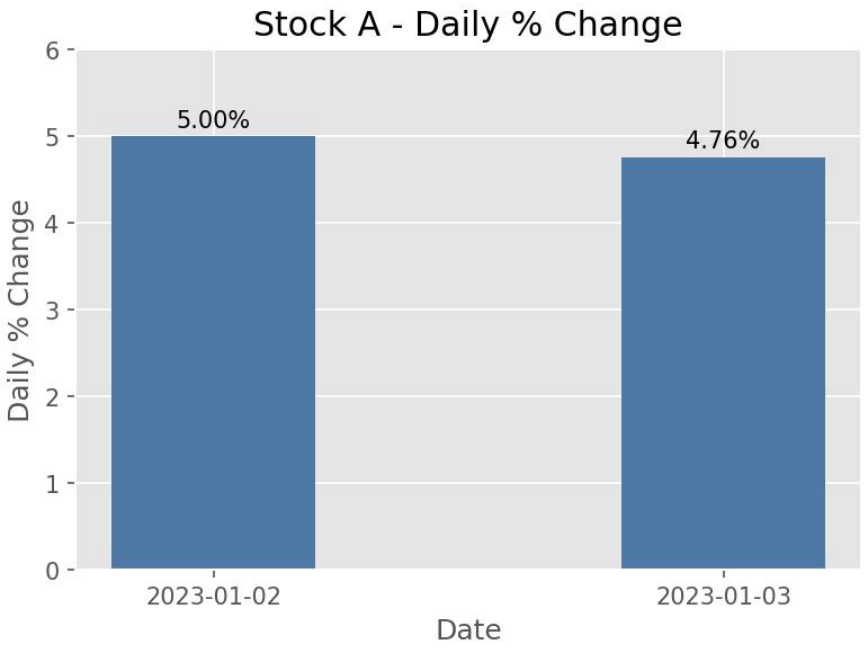



*Figure 20.2: Daily % change of Stock A*

**RESULT**

The daily percentage change of Stock A is 5.00% on 2023-01-02 and 4.76% on 2023-01-03, showing continuous growth with a slightly slower rate on the second day.

## Question 3

Build a table to display the stock price data for each company over the given period. Label the table elements.

**AIM**

To display the stock prices of all three companies for each date in a labelled table.

**PROCEDURE**

1. Select the Date and price columns from the data frame.
2. Print the data and convert it using tableGrob().
3. Display the table with a title.

**CODE**

```
# Table of stock prices
stock_tbl <- stock[, c("Date", "Stock_A", "Stock_B", "Stock_C")]
print(stock_tbl)
tbl <- tableGrob(stock_tbl, rows = NULL)
grid.newpage()
grid.arrange(tbl, top = "Stock Price Data")
```

**OUTPUT**

**Stock Price Data**

| Date | Stock A | Stock B | Stock C |
|---|---|---|---|
| 2023-01-01 | 100 | 150 | 120 |
| 2023-01-02 | 105 | 152 | 118 |
| 2023-01-03 | 110 | 148 | 122 |

*Table 20.1: Stock price data*

**RESULT**

The table gives the prices of Stock A, Stock B and Stock C for 2023-01-01 to 2023-01-03.

## Question 4

Develop a dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

**AIM**

To combine the line chart, bar chart and table into one dashboard for exploring the stock prices.

**PROCEDURE**

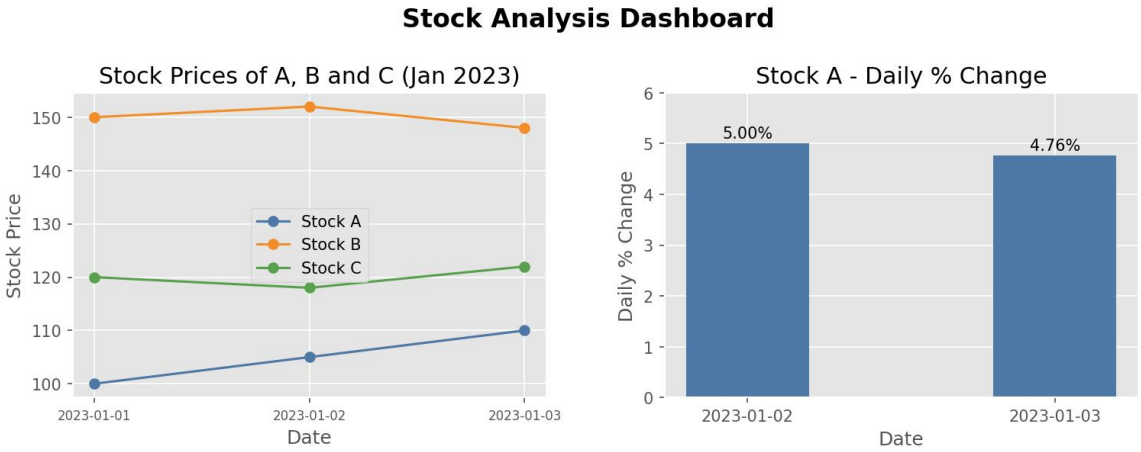
1. Reuse p1, p2 and the table tbl from the previous questions.
2. Arrange them with grid.arrange() under a common title.
3. Use plotly (ggplotly) and DT for interactive exploration.

**CODE**

```
# Dashboard: line chart + bar chart + table
dash <- grid.arrange(
  arrangeGrob(p1, p2, ncol = 2),
  tbl, nrow = 2, heights = c(2, 1),
  top = "Stock Analysis Dashboard")

# Interactive exploration
library(plotly)
ggplotly(p1)
ggplotly(p2)
library(DT)
datatable(stock_tbl, filter = "top")
```

## OUTPUT




| Date | Stock A | Stock B | Stock C |
|---|---|---|---|
| 2023-01-01 | 100 | 150 | 120 |
| 2023-01-02 | 105 | 152 | 118 |
| 2023-01-03 | 110 | 148 | 122 |

*Figure 20.3: Stock analysis dashboard*

## RESULT

The dashboard shows the price trends, the daily % change of Stock A and the price table together, and the interactive versions allow the stock prices to be explored.